

1. Which of the following is NOT an attribute selection measure used in decision trees?

- A) Entropy
- B) Information Gain
- C) Chi-Square
- D) K-Nearest Neighbors (KNN)

Answer: D) K-Nearest Neighbors (KNN)

Explanation:

Decision trees use measures like **Entropy, Information Gain, and Chi-Square** to determine the best split at each node. KNN is a classification algorithm and is not used for attribute selection in decision trees.

2. What is the key difference between a binary split and a multiway split in decision trees?

- A) Binary splits divide the data into two groups, while multiway splits create multiple child nodes.
- B) Multiway splits are used only for numerical attributes, whereas binary splits are for categorical attributes.
- C) Binary splits use entropy, while multiway splits use Gini index.
- D) Multiway splits always result in better accuracy than binary splits.

Answer: A) Binary splits divide the data into two groups, while multiway splits create multiple child nodes.

Explanation:

- **Binary splits** create two branches from a node, dividing the data into **two groups**.
 - **Multiway splits** allow multiple branches, creating **more than two child nodes**.
 - Both binary and multiway splits can be used for numerical or categorical attributes, depending on the decision tree implementation.
-

3. In decision tree pruning, which technique removes unnecessary nodes AFTER the tree has been fully grown?

- A) Pre-Pruning
- B) Post-Pruning
- C) Overfitting Pruning
- D) Random Forest

Answer: B) Post-Pruning

Explanation:

- **Post-pruning** (also called **Reduced Error Pruning**) removes nodes **after** the full tree has been built.
- It evaluates subtrees and removes branches that do not significantly improve accuracy, helping to **reduce overfitting**.
- **Pre-pruning**, in contrast, stops tree growth early based on conditions like minimum samples per split.

4. How is the Chi-Square test used for decision tree splitting?

- A) It calculates entropy to determine the best split.
- B) It measures the statistical significance of differences between parent and child nodes.
- C) It ensures all splits are binary.
- D) It helps reduce the number of categorical features.

Answer: B) It measures the statistical significance of differences between parent and child nodes.

Explanation:

The **Chi-Square test** measures whether a split significantly improves classification by checking differences in observed vs. expected frequencies of target variables. A **higher Chi-Square value** means a better split.

5. What is the main disadvantage of decision trees compared to other machine learning algorithms?

- A) Decision trees are difficult to interpret.
- B) They always underfit the data.
- C) They are prone to overfitting, especially with deep trees.
- D) They require extensive data cleaning.

Answer: C) They are prone to overfitting, especially with deep trees.

Explanation:

- Decision trees **tend to overfit** when they become too complex, learning noise in the training data.
- This issue can be addressed using **pruning** or ensemble methods like **Random Forest** to improve generalization.

MCQs (Decision Tree)

1. Given entropy of parent = 1, weights averages = (3/4, 1/4) and entropy of children = (0.9, 0). What is the information gain?
- a) 0.675
 - b) 0.75
 - c) 0.325
 - d) 0.1

Ans: c)

Explanation: We know Information Gain = Entropy (Parent) – $\sum$ (weights average * entropy (Child)).

$$\begin{aligned}\text{Information Gain} &= 1 - (3/4 * 0.9 + 1/4 * 0) \\ &= 1 - (0.675 + 0) \\ &= 1 - 0.675 \\ &= 0.325\end{aligned}$$

2. Which of the following statements is not true about Information Gain?
- a) It is used to determine which feature/attribute gives us the maximum information about a class
 - b) It is based on the concept of entropy, which is the degree of impurity or disorder
 - c) It aims to reduce the level of entropy starting from the root node to the leave nodes
 - d) It is often promote the level of entropy starting from the root node to the leave nodes

Ans: d)

Explanation: **Information Gain** is based on the concept of entropy and it never tries to promote but tries to reduce the level of entropy starting from the root node to the leave nodes.

3. If a dataset has **three classes** with probabilities **0.2, 0.3, and 0.5**, what is the Gini index?
- a) 0.50
 - b) 0.62
 - c) 0.42
 - d) 0.38

Ans: c)

$$\text{Explanation: Gini} = 1 - ((0.2)^2 + (0.3)^2 + (0.5)^2) = 1 - (0.04 + 0.09 + 0.25) = 1 - 0.38 = 0.62$$

4. The **Gini coefficient** in a decision tree is used to measure:
- a) The depth of the tree
 - b) The impurity of a node
 - c) The number of leaves in the tree
 - d) The accuracy of the model

Ans: b)

Explanation: The **Gini coefficient** in a decision tree is used to measure the impurity of a node. The nodes having the lowest value of Ginni coefficient

5. Which criterion is used by default in **DecisionTreeClassifier()** for classification?

- a) Entropy
- b) Gini
- c) Mean Squared Error
- d) Information Gain

Ans: b)

Explanation: Gini Coefficient is used by default in objective questions on decision tree in python